

Note 3.1 Definition and Examples

Euclidean Vector Spaces

- **Geometric Representation of Vectors in $\mathbb{R}^2$:**

Given a nonzero vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, then $\mathbf{x}$ can be any line segment from (a, b) to $(a + x_1, b + x_2)$.

And its Euclidean length is the length of the directed line segment $\sqrt{x_1^2 + x_2^2}$.

For each scalar α , the product $\alpha\mathbf{x}$ is defined by $\alpha \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \end{pmatrix}$ and the sign of α determine the direction of $\alpha\mathbf{x}$.

And the addition of two vector $\mathbf{u}, \mathbf{v}$ then $\mathbf{u} + \mathbf{v}$ is represented by the directed line segment from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$ when we place the initial point of $\mathbf{v}$ at the terminal point of $\mathbf{u}$. And $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$.

- **Algebraic Representation of the Scalar Multiplication and Addition**
 - For Multiplication:

$$\alpha\mathbf{x} = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}$$

- For Addition:

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and any scalar α .

The Vector Space $\mathbb{R}^{m \times n}$

- **Definition**

Let $\mathbb{R}^{m \times n}$ denote the set of all $m \times n$ matrices with real entries thus the operation will obey certain algebraic rules.

Vector Space Axioms

- **Definition of Vector Space**

Definition

Let V be a set on which the operations of addition and scalar multiplication are defined. By this we mean that, with each pair of elements x and y in V , we can associate a unique element $x + y$ that is also in V , and with each element x in V and each scalar α , we can associate a unique element αx in V . The set V , together with the operations of addition and scalar multiplication, is said to form a **vector space** if the following axioms are satisfied:

- A1. $x + y = y + x$ for any x and y in V .
- A2. $(x + y) + z = x + (y + z)$ for any x, y and z in V .
- A3. There exists an element 0 in V such that $x + 0 = x$ for each $x \in V$.
- A4. For each $x \in V$, there exists an element $-x$ in V such that $x + (-x) = 0$.
- A5. $\alpha(x + y) = \alpha x + \alpha y$ for each scalar α and any x and y in V .
- A6. $(\alpha + \beta)x = \alpha x + \beta x$ for any scalars α and β and any $x \in V$.
- A7. $(\alpha\beta)x = \alpha(\beta x)$ for any scalars α and β and any $x \in V$.
- A8. $1 \cdot x = x$ for all $x \in V$.

• Remarks

- An important component of the definition is the closure properties of the two operations. that is,
 - If $x \in V$ and α is a scalar, then $\alpha x \in V$.
 - If $x, y \in V$, then $x + y \in V$.
- Closure property is that the operation result of the elements in certain set is still in that set.

The Vector Space $C[a, b]$

• Definition

$C[a, b]$ denote the set of all real-valued functions that are defined and continuous on the interval $[a, b]$. And the vectors are the functions in $C[a, b]$. The sum $f + g$ of two functions in $C[a, b]$ is defined by $(f + g)(x) = f(x) + g(x)$. Let α is a real number, then we define αf by $(\alpha f)(x) = \alpha f(x)$. Clearly they are in $C[a, b]$.

• Proof of Vector Space

1. $(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$.
2. $[f + (g + z)](x) = f(x) + (g + z)(x) = f(x) + g(x) + z(x) = (f + g)(x) + z(x) = [(f + g) + z](x)$
3. Let $z(x) = 0$ for all x in $[a, b]$, then $f + z = f$ for all f in $C[a, b]$.
4. Let $\alpha = -1$, then $(\alpha f)(x) = (-f)(x) = -f(x)$, then $[f + (-f)](x) = f(x) - f(x) = 0$ for all x in $[a, b]$.
5. $\alpha(f + g)(x) = \alpha(f(x) + g(x)) = \alpha f(x) + \alpha g(x)$.

6. $[(\alpha + \beta)f](x) = (\alpha + \beta)f(x) = \alpha f(x) + \beta f(x).$
7. $[(\alpha\beta)f](x) = (\alpha\beta)f(x) = \alpha(\beta f(x)) = [\alpha(\beta f)](x)$
8. Let $\alpha = 1$, then $1 \cdot f = f$.

The Vector Space P_n

- **Definition**

P_n denote the set of all polynomials of degree less than n . And define $p + q$ and αp respectively by $(p + q)(x) = p(x) + q(x)$ and $(\alpha p)(x) = \alpha p(x)$. The zero vector is $z(x) = 0x^{n-1} + 0x^{n-2} + \dots + 0$.

Additional Properties of Vector Spaces

- **Theorem 3.1.1**

If V is a vector space and $\mathbf{x}$ is any element of V , then

- $0\mathbf{x} = \mathbf{0}$.
- $\mathbf{x} + \mathbf{y} = \mathbf{0}$ implies that $\mathbf{y} = -\mathbf{x}$.
- $(-1)\mathbf{x} = -\mathbf{x}$.

Exercise Section 3.1

- [Question 16th](#)

Definition

In general, two vector spaces are said to be **isomorphic** if their elements can be put into a one-to-one correspondence that is preserved under scalar multiplication and addition as $\alpha p \leftrightarrow \alpha \mathbf{a}$ and $p + q \leftrightarrow \mathbf{a} + \mathbf{b}$.